

TrES-1b

R-1460

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_200927 · created 2026-08-01T10:39:19+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459119.6849 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.021
Transit depth (Rp/Rs)^2	2.8 +/- 0.71 [%]
Orbital Inclination (inc)	89.77 +/- 0.79
Airmass coefficient 1 (a1)	0.951 +/- 0.011
Airmass coefficient 2 (a2)	0.031 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.19 %
Best Comparison Star	#1 - [565, 268]
Optimal Aperture	3.96
Optimal Annulus	8.75
Transit Duration (day)	0.0858 +/- 0.0023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.021 vs 0.1358 (deviation 1.49σ)
Duration fit vs published	0.0858 d vs 0.1042 d (deviation 7.99σ)
Mid-time O–C	1.4 min
Depth SNR	8.0
Residual scatter	1.19 %

Light curve

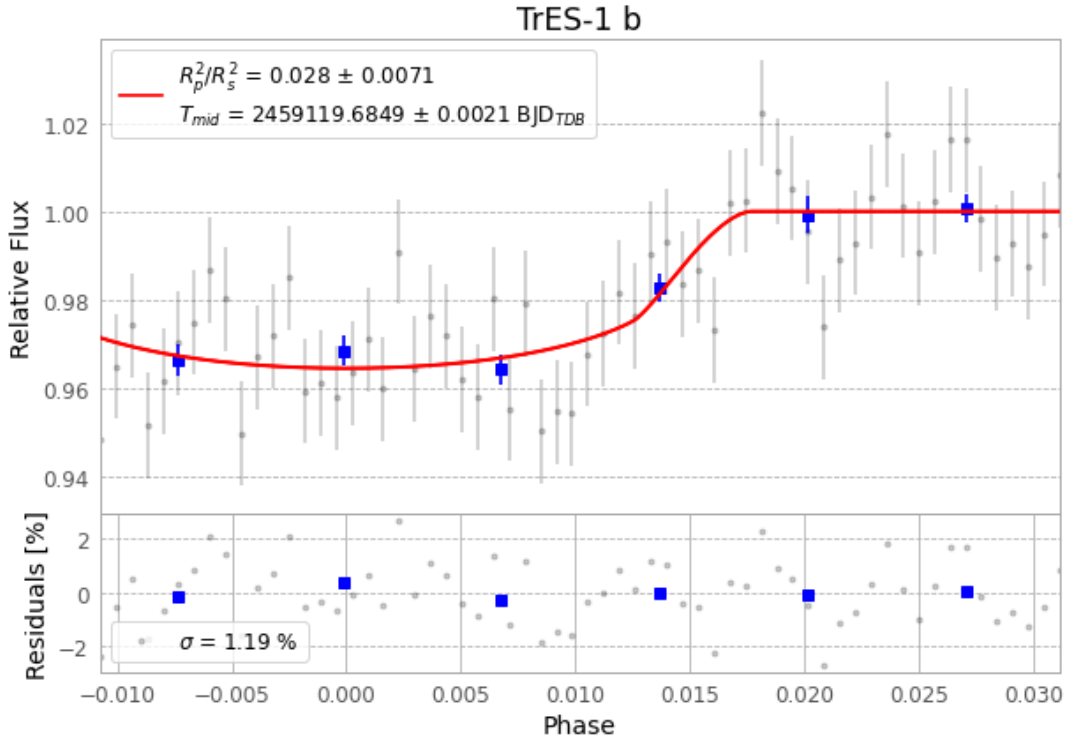


Figure 1 — FinalLightCurve_TrES-1 b_2020-09-26.png (SHA-256 c3b1abf06bb31a6e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [273, 288], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bcdd64f46473964d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

62 FITS frames (41 MB), TRES-1200927033612.FITS → TRES-1200927063913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-200927.log (98aca4efea2f...), FinalLightCurve_TrES-1 b_2020-09-26.png (c3b1abf06bb3...), FinalLightCurve_TrES-1 b_2020-09-26.csv (38c45a4a8adb...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 10:39Z.